Matteo Bernacchia

Tokyo, Japan | work@kikijiki.com | github.com/kikijiki | linkedin.com/in/kikijiki | kikijiki.com

Summary

Software engineer with 10+ years of experience across game development and digital twin platforms, building the foundations other teams depend on: data formats and SDKs, storage and data pipelines, 3D viewers, React UIs, and CI/CD, mostly in Rust, Python, and TypeScript, from design through production release. Hands-on experience running workloads on AWS.

Currently building core platform systems, from viewers and the data layer to computer vision pipelines for vehicle detection and tracking, with security hardening and compliance for platform services, and working with solutions engineers to deliver for factory customers in Japan and overseas. Design and build AI agents end to end (RAG, tool calling, tracing, evals), with an understanding of AI from the underlying math to current techniques. Beyond work, build substantial AI systems of my own, such as a from-scratch LLM inference engine.

Work Experience

Senior Software Engineer – Digital Twin Platform Oct 2025 — Present
Woven by Toyota *Tokyo, Japan*

- Led the Visualizer and Data Pipeline teams (6 people in total): requirements gathering, coordination with other teams, task and schedule management, reviews and planning, security audits
- Work with solutions engineers to gather needs directly from multiple factories (external companies) in Japan and overseas and feed them into development
- Built a modular viewer engine (data components, logic, signals, rendering) with plugins for 3D models, Gaussian splats, shapes, text, and images, plus a React panels API with ready-made timeline, annotation, inspection, and profiling panels; adopted by multiple teams to build their own visualizers
- Improved rendering performance for large files and heavy scenes, and playback and seek performance through data streaming and snapshot support; improved documentation and templates and hardened security
- Continued evolving the data format and SDK with spec improvements and advanced features including snapshots, sharding, and async I/O; improved dataframe/Databricks integration
- Implemented a graph-based object storage and data layer that serves every service on the platform and makes onboarding new services fast; supports deduplication, versioning, tags, queries, and recursive operations, backed by an extensive test and fuzzing suite
- Ported a C#/Unity simulation engine to Rust (Bevy), eliminating licensing fees and improving execution speed by 100x, memory usage by 50x, and binary size by 20x
- Built computer vision pipelines for vehicle detection and tracking: auto-labeling with SAM 3.1, training RF-DETR detectors, a fusion algorithm merging tracks from multiple cameras, and tooling for experimentation
- Developed a fully working tech demo presented at Toyota's public Kakezan showcase in Woven City, combining project management, simulation editor and execution, visualizers, KPI dashboards, and an integrated AI agent
  - Agent present throughout the multi-tool, multi-panel app, working seamlessly with the editor, visualizers, and dashboards; RAG, telemetry DB queries, and file/data search integrated as tools
  - LLMs via AWS Bedrock; OpenTelemetry tracing of turns and tool calls; evals measuring accuracy on common tasks to improve prompts and context design

Senior Software Engineer (Contractor) – Digital Twin Platform Aug 2024 — Sep 2025
Woven by Toyota *Tokyo, Japan*

- Designed a unified data format specification, now the core technology used across the Digital Twin Platform for telemetry data streaming and storage; implemented the SDK with a Rust core and bindings for C#, Python, and WebAssembly (JavaScript/TypeScript); added CLI tools and dataframe support; integrated with viewers, simulation engines, and data pipelines
- Developed a React SPA (TypeScript/Three.js/WebGL) for 3D simulation-data playback, with a 3D viewport, UI overlays, frame and asset streaming, tests, and CI/CD; met all security requirements and shipped
- Added real-world telemetry playback with time-series streaming, asset animation, and custom visualizations; contributed to the design of the mapping from telemetry to viewer entities and animations
- Designed a data model based on an ECS-style architecture where components are the semantic unit of data; implemented the component registry, plus related tooling, documentation, and frontend/backend support
- Coordinated GitHub migrations for multiple projects; implemented CI/CD pipelines for build, test, benchmarking, audit, SBOM scanning, publishing, documentation websites, and releases

Lead Tools Programmer Sep 2018 — Jul 2024
Square Enix *Tokyo, Japan*

- Led a five-person team developing a company-wide asset platform with ML-powered tagging and captioning
- Designed a scalable backend using Rust, PyTorch, and PostgreSQL, and built the web interface with Axum and Leptos
- Developed a visual issue-tracker plugin for Unreal Engine, adopted by three projects (150–300 users total)
- Created a flexible Jenkins CI/CD template for Unreal Engine projects, reducing setup time by weeks per project
- Served as technical project manager for a procedural cutscene generation project; pioneered the adoption of Rust for tools
- Led tools and infrastructure development for a 50-person tech demo team, covering Perforce setup, Jenkins CI, server administration, and Sentry integration
- Mentored engineers, managed task allocation, handled stakeholder requests, and collaborated with cross-functional teams

Game Programmer – Elden Ring Apr 2015 — Sep 2018
FromSoftware *Tokyo, Japan*

- Implemented the full vegetation system pipeline: a 3ds Max DCC tool, BVH exporter, runtime streaming, and rendering
- Converted the AI system from single-threaded to multi-threaded execution and improved Havok pathfinding for the open world
- Created open-world map-building tools that merge assets, events, and metadata across LODs
- Led the Git migration and GitLab deployment for the engine repositories

Programmer (Internship) Apr 2014 — Oct 2014
Cygames *Tokyo, Japan*

- Developed the mobile app "O'REILLY COLLECTION" using Unity and Adobe AIR
- Implemented a custom Yammer-based chat client for internal use
- Received a commendation from the department lead for outstanding performance

Publications

Believable Fighting Characters in Role-Playing Games Using the BDI Model Feb 2015

- Matteo Bernacchia, Jun'ichi Hoshino – IPSJ – cir.nii.ac.jp/crid/1573950402597012736

AI Platform for Supporting Believable Combat in Role-Playing Games Nov 2014

- Matteo Bernacchia, Jun'ichi Hoshino – IPSJ – cir.nii.ac.jp/crid/1050574047113723520

Skills

- **Languages**: Rust, C++, Python, TypeScript/JavaScript, C#, SQL
- **Engines & Frameworks**: Unreal Engine, React, Three.js, Node.js, DirectX/Vulkan, CUDA
- **AI & ML**: LLMs/GenAI (Amazon Bedrock), AI agents (tool calling, workflows), MCP, ACP, RAG, prompt/context engineering, evals, PyTorch, MLOps
- **Tools & Infra**: AWS (EKS), Git, Perforce, GitHub Actions, Jenkins, Docker, Kubernetes, Terraform, OpenTelemetry, Nix, Linux (NixOS)
- **Spoken Languages**: Italian (Native), English (Professional; TOEFL iBT 110, TOEIC 940), Japanese (JLPT N1)

Certifications

- **Deep Learning & AI**: Deep Learning, MLOps, NLP, GenAI, Reinforcement Learning (DeepLearning.ai)
- **Cloud & Systems**: AWS Cloud Solutions Architect Professional Certificate, Rust Programming, GPU Programming, PostgreSQL (Coursera)
- **Full-Stack, Data & Security**: Meta Front-End & Back-End Developer, Google Advanced Data Analytics, Google Cybersecurity (Coursera)

Education

University of Tsukuba (Graduate School) Apr 2013 — Mar 2015
Master's Degree, Systems and Information Engineering, Intelligent Interaction Systems *Tsukuba, Ibaraki, Japan*

- Researched believable AI characters for role-playing games, based on the BDI (belief–desire–intention) agent model

Marche Polytechnic University Jul 2010
Bachelor's Degree, Information and Automation Engineering *Ancona, Marche, Italy*

- Thesis: Study of a Software Agent Platform for Healthcare Information Systems (2010, Faculty of Engineering)